

MLFF current multi-view target-subset chain specification

Status: current normative FEAS1/MVIDX1/MVSEL2/REPAIR2/MVSTATE2/MVQUAL integration contract

Architecture: revision 105

1. Scope

This specification owns the current integrated multi-view target-subset construction chain after DATA7 fitted preparation. It defines the exact scientific relationship among FEAS1, MVIDX1, MVSEL2, REPAIR2, MVSTATE2, and MVQUAL and the current forward/lazy execution invariants.

Target-size screening after prefix qualification is owned by `mlff_target_subset_size_study_spec.md`.

There is no alternate current MVSEL1, REPAIR1, MVSTATE-REUSE1, migration, or rescue path.

2. Scientific input contract

For each canonical DATA5 fold/final gradient-training domain, the chain consumes one current DATA7 TargetSubsetInputBundle containing the fitted feature/metric, weights, conditions/provenance/correlation identities, hard obligations, representative/diversity evidence, event/environment/focus evidence, and training-domain difficulty inputs required by the selector policy.

No held-out CV, calibration, or locked-test evidence may enter these fitted inputs or target membership.

Scientific calculations use FP64 decision arithmetic unless a narrower current specification explicitly defines an exact-equivalent representation.

3. Exact hard-coverage contract

For feature family m , candidate c , witness w , fitted scaling D_m , and frozen witness radius $r_w(m)$, exact adjacency is

$$A[m, w, c] = 1 \text{ iff } ||D_m (x_w(m) - x_c(m))||_2 \leq r_w(m)$$

The current default hard family coverage threshold is exactly:

0.95

unless an explicitly identified material/profile policy defines another threshold for a named family. Any such override is part of scientific policy identity.

The current default floating contender tolerance for the selector/repair policy is exactly:

1e-14

and every best-relative floating filter uses the inclusive form

`value >= best - tolerance`

under the canonical FP64 reduction semantics.

Candidate, witness, family, obligation, correlation-unit, and stable UID ordering are scientific identity.

4. FEAS1

FEAS1 evaluates the complete eligible candidate/reference pool before progressive selection. It SHALL:

- validate exact expected support/self-support as required by the current coverage policy;
- measure candidate/witness support and fragile low-support witness mass;
- evaluate mandatory hard-obligation feasibility;
- identify capacity limitations relevant to the nominal target-size population;
- preserve candidate/reference/family/policy identities in its evidence.

FEAS1 does not relax hard coverage or invent candidate sizes when support is insufficient.

5. MVIDX1

MVIDX1 owns one authenticated exact sparse relation for the semantic neighborhood inputs. Its scientific identity binds at least:

- domain/candidate/reference identities and canonical ordering;
- family identities and fitted scaling/distance/radius semantics;
- exact typed sparse arrays/cardinalities;
- hard-obligation incidence and correlation-code lineage where packaged with the runtime projection;
- current schema/cache semantic version.

MVIDX1 may persist witness-oriented and candidate-oriented sparse views. The MVSEL2/REPAIR2 runtime consumes only the candidate-to-witness, candidate-to-obligation, and correlation-code projection required by the current algorithm.

Worker count, chunk size, queue order, in-memory versus file-backed inversion, mmap path, and similar execution choices are non-semantic when the authoritative sparse content is identical.

6. MVSEL2 sole ordering authority

MVSEL2 produces one deterministic progressive target order per canonical training domain.

6.1 Hard-coverage phase

During the hard-coverage phase, candidate comparison follows the current staged exact lexicographic authority:

1. maximum hard gain;
2. first canonical minimum-coverage/bottleneck family;
3. best relative bottleneck-family gain;
4. total coverage gain;
5. least-selected correlation-unit balance;
6. harmonic representative gain;
7. sparse diversity;
8. stable UID.

An implementation MAY apply exact staged scans/filters so long as every eliminated candidate is provably unable to win under the same lexicographic policy and inclusive tolerance semantics.

6.2 Representative phase

Once hard coverage is satisfied, the current representative-fill authority orders contenders by:

1. representative gain;
2. correlation balance;
3. diversity;
4. stable UID,

with the same current tolerance/tie semantics.

6.3 Forward/lazy execution

MVSEL2 uses candidate-forward sparse rows rather than complete eager inverse candidate-marginal propagation.

After the hard phase, the implementation performs an exact all-candidate rebase before constructing the certified lazy representative frontier. Cached stale scores are outward-rounded conservative upper bounds. A candidate may remain unrefreshed only while its bound proves it cannot beat the best exact contender under the frozen tolerance. Otherwise it is refreshed exactly.

Full-forward exact scoring is the correctness oracle and bounded fallback. Lazy/frontier scheduling does not change scientific rank authority.

The selected order SHALL be invariant under qualified worker count, batch size, queue order, cache location, restart point, or frontier-rebuild schedule.

7. REPAIR2 sole repair authority

REPAIR2 consumes the MVSEL2 order and current forward state. It preserves:

- active-shell-only repair;
- immutable protected lower prefixes;
- zero-unique/hard-safe removal admission;
- exact deficit/frontier replacement scoring;
- hard-obligation safety;
- strict no-hard-coverage regression;
- current objective/tolerance/tie hierarchy;
- rank inheritance and deterministic future displacement;
- bounded passes/swaps/shortlist behavior as defined by current repair policy;
- deterministic accepted/rejected repair trace.

Hypothetical proposal scoring may execute concurrently within one immutable pre-swap state. Authoritative winner comparison and accepted state mutation remain deterministic.

REPAIR2 publishes one repaired master order per domain. Candidate target sizes are prefixes of this single order; independent per-rung repair is prohibited.

8. MVSTATE2 current continuation state

Current continuation-state records use the MVSTATE2 family. A state record binds at least:

- dataset/domain identity;
- canonical candidate UID and family order;
- DATA7/FEAS1/MVIDX1 identities;
- witness weights and hard obligations;
- correlation-unit identities;
- selector/repair policy identity;
- exact selected prefix/order state;
- current schema/version identity.

The durable compact payload contains the exact continuation quantities required by the current algorithm, including witness multiplicity, family covered mass, obligation counts, correlation counts, and representative utility/state.

Complete per-candidate marginal arrays and ephemeral lazy-heap contents are not durable authority.

Publication SHALL be transactional/authenticated. Restoration SHALL validate manifest/payload integrity and recompute continuation invariants from the selected prefix before use.

An artifact that does not validate as current MVSTATE2 is rejected/reconstructed or requires campaign re-preparation. Historical state is not migrated into current authority.

9. MVQUAL independent qualification

MVQUAL independently recomputes the hard coverage/obligation evidence for required repaired prefixes using authenticated primitive sparse inputs.

It SHALL NOT accept MVSEL2/REPAIR2 internal counters as independent qualification evidence.

A qualification record binds the requested domain/prefix size and current identities and reports, as applicable:

- per-family coverage and deficit;
- uncovered weighted mass/count;
- mandatory-obligation satisfaction;
- redundancy and unique-support diagnostics;
- provenance/correlation diagnostics;
- current pass/fail result.

MVQUAL completion order may be parallel/non-authoritative; persisted result ordering and scientific predicates are deterministic.

10. Nested-prefix invariant

For repaired master order $\text{pi}[d]$,

$D[d, N] = \text{pi}[d][:N]$

and for $N_2 > N_1$,

$D[d, N_1]$ is a strict prefix/subset of $D[d, N_2]$.

Under the fixed positive hard coverage/obligation predicates, hard satisfaction cannot regress as N grows. A pass/fail/pass qualification sequence is a hard invariant failure and SHALL stop target-size preparation.

11. Resource contract

The chain SHALL be realizable at target scale with one product-scale sparse/selection authority per semantic domain identity rather than one copy per target-size rung.

Current execution MUST NOT require:

- witness-to-candidate inverse arrays to be mapped by MVSEL2/REPAIR2 normal execution;
- complete per-candidate marginal arrays as durable selector state;
- an independent descriptor/MVIDX/selector/repair copy for each nominal target size.

Out-of-core sparse inversion, mmap/file-backed arrays, chunking, queue backpressure, and reconstructible caches are permitted exact execution realizations under their current runtime specifications.

12. Qualification requirements

A release implementation of this current chain requires evidence covering, as applicable:

- exact MVIDX sparse-array validation;
- deterministic order/repair equality across qualified worker/batch/restart/frontier schedules;
- lazy-versus-full-forward oracle equality;
- MVSTATE2 corruption/staleness/restart checks;
- independent MVQUAL equality to its current predicate definitions;
- nesting and hard-coverage monotonicity over the nominal/materializable prefixes used by the size-study policy;
- bounded RAM/scratch/restart behavior at representative target scale;
- performance evidence demonstrating that the exact forward/lazy realization is suitable for production-scale selection.

Performance qualification does not authorize a scientific-policy change, and GPU qualification is not implied by CPU/sparse-chain qualification.

13. Unsupported historical artifacts

Current new-campaign schemas/identities are the current MVSEL2/REPAIR2/MVSTATE2/MVQUAL families defined by the implementation specifications. Old MVSEL1, REPAIR1, MVSTATE-REUSE1, generated-size rescue, and migration records are non-current historical evidence only.

There is no requirement that current readers preserve their construction/runtime semantics. If an old campaign cannot validate against the current generation, the user must re-prepare it.